

The *mélange* of Islamic, Jewish, and Christian contributions to astrological and astronomical debates in the Middle Ages was particularly complex and manifold in Spain between the tenth and the twelfth centuries. While this feature is increasingly acknowledged in recent research, not many scholars address the period before the tenth century. Such a lacuna can lead to the assumption that only the schools of translators in Spain and the cultural contacts of that epoch brought those regions that were dominated by Christian culture into contact with philosophy and science. This, however, is a misperception. Therefore, I have to begin my overview of Christian astrology in the early Middle Ages.

Sharing Muslim Knowledge: Christian Astrology

Quite contrary to a widely held assumption, astrology continued to flourish in medieval Christian contexts as well.²⁹ Not only did scholars of the time try theologically and philosophically to find a possibility for distinguishing ‘permissible’ from ‘prohibited’ astrology; politicians, too, time and again supported astrology and used it for legitimating their power, exactly as the Roman emperors had done before.³⁰ This is true, for instance, of Charlemagne, Louis the Pious, Henry II, and the Staufer Emperor Frederick II.

Especially in the Byzantine East, within ‘Greek Christianity,’ we have to assume a living tradition of astrological doctrines that were, of course, transformed according to Christian myths and theological positions.³¹ The history of ‘Latin Christianity’ in the West was somewhat different. Here, the writings of Firmicus Maternus in particular were repeatedly copied until the high Middle Ages. Slowly, the increasing exchange with the Arabic cultures of Europe and the translation of astrological texts into Latin led to an adaptation of more learned scientific astronomy and astrology.

The Roman senator Anicius Manlius Severinus Boethius (c. 480–524), who led a tragic career under the eastern Gothic king Theodoric

²⁹ On late antiquity see von Stuckrad, *Das Ringen um die Astrologie*, 534–800. On the subsequent history of reception see Hübner, *Zodiacus Christianus*; von Stuckrad, *Geschichte der Astrologie*, 185–206. Cf. also Blume, *Regenten des Himmels*.

³⁰ Stierlin, *L’astrologie et le pouvoir*, especially chapters 5 and 7.

³¹ An overview is provided in Gundel & Gundel, *Astrologumena*, 213–254.

the Great, was important for subsequent generations. Shortly before his death in the dungeon of Pavia, Boethius wrote his most influential work, the *Consolatio philosophiae* ("Consolation of Philosophy"). In the form of a dialogue that he has with the "Lady Philosophia," Boethius here presents once more the complete ancient tradition and thus conveys it to the Christian world.³² With regard to astrology this was important insofar as Boethius strengthened Neoplatonism and also provided important arguments for the discussion of providence, determinism, fatalism, and free will.

Boethius' influence was significantly superseded by the impact of Isidore of Seville (c. 570–636) who worked under the western Goths and became an important point of reference for subsequent Christian opinions concerning astrology. From 601 until his death, Isidore was bishop of Seville. He penned a large number of works that addressed all sorts of knowledge important in his day. In order to improve the educational level of the western Gothic court, Isidore collected ancient pieces of knowledge and combined them in a twenty-volume encyclopedia,³³ known under the title *Origines* or *Etymologiae*, because his main method of explanation was a (more or less reasonable) etymological derivation. Besides the *artes liberales*, medicine, jurisprudence, theology, philosophy, social sciences, anthropology, zoology, physics, geography, architecture, minerals, agriculture, warfare, theater, clothing, handwork, household equipment, and other things, the learned bishop also addressed astrology and astronomy. Although delineated differently on an etymological basis, Isidore—like all of his contemporaries—saw those disciplines as two aspects of the same interest. Another work (*De rerum natura*, "On the Nature of Things") that engaged cosmology and astrology he dedicated in 613 to King Sisebut, who himself had written an astrological didactic poem.

Isidore distinguishes between *astrologia superstitiosa* and *astrologia naturalis*. The first, insofar as it intends to predict the character and fate of an individual, is regarded as superstition that has become

³² The manuscripts were widely spread in the Middle Ages. King Alfred (d. 901) translated the work into Anglo-Saxon, the monk Notker Labeo (d. 1022) into German, and Maximus Planudes (d. 1310) into Greek.

³³ In fact, this work remained incomplete, and the twenty-volume edition was prepared by Isidore's friend Braulio, bishop of Saragossa, who also was initiator and addressee of the work. The *Etymologiae* are usually filed among the most important transmitters of ancient knowledge into the Middle Ages, along with Martianus Capella's *De nuptiis Mercurii et Philologiae* and Cassiodorus' *Institutiones*.

superfluous due to the birth of Christ. As can be seen from his explanation of the ‘Star of Bethlehem,’ however, this did not mean that astrology was wrong—it simply was illicit, which is a big difference. Thus, Jim Tester notes: “The idea, at least, of a potentially valid science of astrology was kept alive by the very authorities who condemned it.”³⁴ From the reading of individual horoscopes Isidore differentiated *astrologia naturalis*, which tries to scientifically fathom the nature of things and is by no means at odds with Christian doctrine. Among these accepted branches of astrology he listed meteorological and, in particular, medical astrology. He even claimed that every physician should have an astrological education, a claim that was to gain wide acceptance up until the Renaissance. It certainly is no exaggeration when we say that Isidore of Seville with this distinction provided an interpretational framework for Christians that ultimately led to the acceptance and increased practice of astrology from the eleventh century onward.

The history of reception of Boethius and Isidore began right after their demise. Already Bede (673–735), whom later generations called Beda Venerabilis (“the Venerable Bede”) and whom the Roman Catholic Church in 1899 declared the only English doctor of the church, built on their teachings and used them for his chronological calculations that defined the Christian festal calendar until the Renaissance. We should not forget that dividing and measuring time was a difficult task because precise instruments for determining the sun’s positions were scarce. Charlemagne thus must have been delighted when in 807 the caliph Hārūn al-Rashīd gave him an elaborate water clock that could exactly measure hours of equal and unequal length. Charlemagne was highly interested in astronomy and astrology, as we know from his correspondence with Alcuin of Tours, his friend and teacher. His court scribe Einhard, too, repeatedly attested to Charlemagne’s interest in astronomical observation. Therefore, it is not surprising that this king—and after him his son Louis the Pious, about whom it was said that he had astrologers around him all the time—continuously supported the educational system of his empire. Already before Charlemagne, missionaries had founded monastery schools in Fulda, St. Gallen, Reichenau, and Regensburg, and from the ninth century

³⁴ Tester, *History of Western Astrology*, 126.

on, interest in a good education of the clergy grew. The monks in these schools were not only engaged in theological, philological, and philosophical disciplines, but also in agriculture, music, and astronomy. The latter was, as noted, especially important for calendrical issues.

The genre of the *computus*, a system for calculating calendars developed by Bede, found wide acceptance and was soon supplemented by country sayings and astrological rulings that people took from Firmicus Maternus or from other sources. Our modern word “computer” is derived from *computus*, another indication of the fact that much of our modern concept of time stems from medieval elaborations of ancient considerations.³⁵ An enormous task of the ninth century was the calendrical reform of Charlemagne whose ‘imperial calendar’ remained valid until the thirteenth century.³⁶ Of this calendar there are more than fifty manuscripts in eight different versions extant, stemming from all over Europe. There are Romanic, Germanic, and Celtic versions (a Slavic has not yet been found); the oldest—Rhine-Frankish—version was written in 789 in the abbey of Lorsch.

Therefore, as early as the early ninth century, a transfer of knowledge from Muslim to Christian cultural domains took place, and Christians combined this knowledge with their own tradition. One century later—thus before the great translation projects from Arabic into Latin which began later—monks at the Benedictine monastery of Santa Maria de Ripoll, which lies before the Pyrenees, collected treatises on arithmetic, geometry, astronomy, and calendrical issues, the translations of which soon circulated all over Europe. For a long time, historians did not understand how Hermann of Reichenau, also known as Hermann the Lame (1013–1054), in his remote valley of the Austrian Alps could write a detailed treatise on the use of the astrolabe. The answer is simple: he had access to an edition of the Ripoll texts.³⁷ Hermann, an excellent mathematician, published a critique of Bede’s calculations in 1040 and synchronized the Easter calendar, so important for Christians, with the actual mathematical data. In his work *De mensura astrolabii* (“On Measuring with the Astrolabe”),

³⁵ See Borst, *Computus*.

³⁶ An excellent edition was published by Borst (ed.), *Der karolingische Reichskalender*; see also his sketch of the research history in Borst, *Die karolingische Kalenderreform*, 1–29.

³⁷ North, *History of Astronomy and Cosmology*, 205.

written around 1045, Hermann reorganized the Arabic lunar calendar of his Spanish sources into the Latin, Julian, solar calendar.³⁸

With Hermann we have reached a period that was characterized by a lively process of cultural encounters between Muslims and Christians. Let me illustrate this exchange between Greek-Arabic science and Christian culture with Gerbert of Aurillac (c. 940–1003). The career of this scholar is representative of the clergy's high interest in scientific innovation in the tenth century.³⁹ As a novice in Barcelona, Gerbert had already been educated in the free arts; he learned Arabic, mathematics, arithmetic, and music, and was one of the first to bring that knowledge across the Alps. Soon his erudition became famous and impressed the pope in Rome. In 983 he was appointed abbot of the monastery in Bobbio and in 991 archbishop of Rheims (even if first without papal acknowledgment). While some people were hostile to him because of his negligible pedigree, it was very fortunate for his career that he was the teacher of Emperor Otto III. Pope Gregory V (996–999), Otto III's cousin, appointed him archbishop of Ravenna in 998. The Emperor elected him to succeed Gregory V as pope in 999. Gerbert took the name of Sylvester II, alluding to Pope Sylvester I (314–335), the advisor of Emperor Constantine I (324–337).

In his writings, Gerbert stresses how important it is to absorb Arabic science for the proliferation of Christianity. Due to his influence he succeeded in spreading astronomical knowledge in monastery schools and the large European centers of learning. Referring particularly to the astrolabe texts of Ripoll, he promoted the introduction of Arabic terminology and the translation of relevant texts into Latin. This policy made it possible for other Christian scholars—Fulbert of Chartres, Hermann of Reichenau, or Walcher of Malvern—to be studied intensively. “This was a time when Hindu-Arabic numerals were slowly finding favour among astronomers in Europe. The change was far from sudden.”⁴⁰ Certainly, for Gerbert the science of the stars occupied the first position among the four classic arts, the so-called Quadrivium of arithmetic, music, geometry, and astronomy.⁴¹

In his work on the astrolabe Gerbert also addressed astrology. In the same vein as Isidore of Seville before him, the learned pope repudiated

³⁸ Borst, *Die karolingische Kalenderreform*, 329–334.

³⁹ See Lindgren, *Gerbert von Aurillac*.

⁴⁰ North, *History of Astronomy and Cosmology*, 206.

⁴¹ Lindgren, *Gerbert von Aurillac*, 38–39.

the interpretation of birth charts as illicit superstition, but he clearly advocated “natural astrology” as an important element of natural science. Presumably, Gerbert had already used the astrolabe in 989 at the cathedral school of Rheims. It is not difficult to reconstruct the way in which the knowledge of using the astrolabe spread from there and other places—for instance the monastery of Reichenau—to the whole of Christian Europe in the eleventh century, and how this instrument soon belonged to the basic technical equipment of Christian astrologers. The astrolabe made observation of the sky much easier, because with this instrument it is sufficient to measure the angle of a single, well perceivable star, and calculate its position. The positions of all other fixed stars could then simply be read from the astrolabe. In this way it was easy to determine the ascendant, which is a crucial prerequisite for calculating horoscopes. In the beginning, however, its usefulness for calculating time and calendar was given priority.

The second half of the eleventh century saw an intensive occupation of Christian scholars with Arabic astronomy and astrology. With translation projects, and later also with their own research, Christians tried to catch up with the progressive knowledge of Muslims and Jews, and both upon the terrain of literature and the terrain of art and iconography a significant rise of astrological motifs can be observed.⁴² The crusades had their share, too, in intensifying processes of cultural exchange,⁴³ with the result that from the twelfth century at the latest we have to talk of a pan-European tradition of knowledge and science. It was centers such as the School of Chartres that first adopted Muslim doctrines and combined them with Christian theology. The theologian Peter Abelard (c. 1100–1140) was affiliated with this school too. He became known for his scholastic dialectic, which he of course applied to astrology as well. He argued that astrology indeed was able to document and predict the *naturalia*, namely the natural causes of changes in agriculture or medicine, but not the *contingentia* that are dependent on chance and the will of God. This interpretation—an elaboration of Isidore’s distinction—was shared by Hugo of St. Victor (died 1141) who lectured at the Victorine monastery school in Paris. It determined the official church’s position for centuries.

⁴² See Blume, *Regenten des Himmels*, 18–63.

⁴³ Mayer (ed.), *Kreuzfahrerstaaten als multikulturelle Gesellschaft*.

A contemporary of Peter Abelard and Hugo of St. Victor was the famous scholar Adelard of Bath who worked at the court of King Henry I of England. Although many details of his biography are still unknown, there can be no doubt that Adelard played a central role in the transmission of Arabic science and the new flourishing of astronomy and astrology in Christianity.⁴⁴ He did not restrict his studies to pure reception but carried out his own research in which he systematically interpreted the information he had collected on his travels to Sicily, Syria, and presumably also to Spain. He translated the tables of al-Khwārisīmī and speculated, referring back to Abū Ma'shar, on the rise and demise of world empires and religions. He explained the difference between the Christians on the one hand, and Jews and Muslims on the other, by the fact that Christians were under the influence of sun and Jupiter, while Muslims and Jews were influenced by Saturn, Mars, and Venus. One reason for this was that Muslims sanctified Friday (Venus), the Jews Saturday (Saturn), and the Christians Sunday (sun). Such speculations were picked up by other thinkers. Joachim of Fiore (died 1202), for instance, predicted in his *Evangelium aeternum* for the year 1260 the dawning of a new age of the Holy Spirit and the establishment of an entirely spiritualized church. The Lateran Council of 1215 was constrained to declare such positions, which spread rapidly, as heretical.

By the end of the twelfth century, many important works of antiquity and their Arabic successors were available in Latin translation. John of Seville, Gerard of Cremona, Plato of Tivoli, Robert of Chester, Hermann of Dalmatia, and others had contributed to a thorough renewal of the sciences in European centers of learning, a renewal for which Charles Homer Haskins coined the expression "Renaissance of the twelfth century." Although nearly all translators were clerics, the cathedral schools, particularly in France, were swamped by aspirants, and famous teachers drew students from all over Europe. The result was a dissemination of these fields of knowledge beyond the clerical domain into circles of court and nobility.

Hence, what can be termed the European Renaissance had been already prepared in this period. Absorbing ancient philosophy and

⁴⁴ See already Haskins, *Studies in the History of Medieval Science*, 20–42; for more recent research see Burnett (ed.), *Adelard of Bath*, with Burnett's own chapter on "Adelard, Ergaphalau and the Science of the Stars"; Cochrane, *Adelard of Bath*.

science through the Muslim transmission changed the attitude of the observing human being vis-à-vis the cosmos. Nature as a this-worldly object became the center of interest, which at the same time led to nature's desacralization and subjugation to human control. Astrology thus played a significant role in the formation of the modern natural sciences.⁴⁵

At this point I conclude my overview. Many other examples could have been mentioned, for instance the scientific activities of the Staufer Emperor Frederick II and the scholar Michael Scot who worked at his court.⁴⁶ But the overall conclusion would be the same: despite theological critique, astrology remained an accepted science in the Middle Ages that was practiced and shared beyond religious borders. We can also conclude that the image of the Middle Ages as an era of brute violence and a lack of 'science' and 'enlightenment' is a misleading oversimplification. The example of astrology shows that in the ninth century a peaceful transfer of knowledge between Muslims, Christians, and Jews had already begun; paradigms of reason and rationality, as well as philosophical issues, belonged to the shared heritage of these religious communities, notwithstanding the difference in interpretation due to religious rhetoric and interest.

If this is true, we will have to challenge accepted periodizations of European history. The characterization of the Renaissance as the birth period of enlightened modernity is today more doubtful than ever. On closer investigation, many of the 'revolutions' between the fifteenth and seventeenth centuries turn out to be a continuation of medieval thought. At the same time, it cannot be denied that the early modern period saw an enormous cultural change in the area of technical development. One change that is of particular relevance for us here is what Elizabeth L. Eisenstein called "the printing revolution in early modern Europe." Indeed, more than twenty-five years after the publication of Eisenstein's work, it is still true that the

transformation of occult and esoteric scribal lore after the advent of printing [...] needs more study. Some arcane writings in Greek, Hebrew or Syriac, for example, became less mysterious. Others became more so. Thus hieroglyphs were set in type more than three centuries before their

⁴⁵ See von Stuckrad, *Geschichte der Astrologie*, 199–206, 213–226, 525–274.

⁴⁶ See von Stuckrad, *Geschichte der Astrologie*, 196–199; on Michael Scot's *Liber introductorius* see also Blume, *Regenten des Himmels*, 52–63; Campion, *History of Western Astrology*, vol. II, 54–56.

decipherment. These sacred carved letters were loaded with significant meaning by readers who could not read them.⁴⁷

In a dialectic between hiding and revealing primordial knowledge, the role of the printing press should not be underestimated. Being a major tool in inter-confessional polemics of the Reformation, the dissemination of esoteric treatises—even more widely than new scientific treatises—was reinforced by printing; it fostered a new trend to ‘arcanization.’ Thus, “one should not think only about new forms of enlightenment when considering the effects of printing on scholarship. New forms of mystification were encouraged as well.”⁴⁸

This is an apt reminder of the fact that the discursive changes that took place between the fifteenth and the seventeenth centuries should be analyzed in their respective scientific, social, political, religious, and cultural contexts.

⁴⁷ Eisenstein, *Printing Press as an Agent of Change*, 77.

⁴⁸ *Ibid.*, 77–78.